Chapter 3 QP + MS Screenshot Pack

Cambridge O Level Physics 5054 — Waves and Light up to 3.2.2

This version uses screenshot pages from the question papers in the repository PDF and pairs them with the respective Paper 2 mark-scheme extracts. The selected questions contain more than 30 exam subquestions and cover 3.1, 3.2.1 and 3.2.2.

Area	Covered by
Wave motion/features/equation	Sections 1–3, 6–7
Reflection/refraction/diffraction in ripple tank	Sections 1–3, 6–7
Light reflection and plane mirrors	Section 4
Light refraction, refractive index, critical angle/TIR, optical fibres	Sections 3–5

1) Waves: longitudinal waves, ripple tank refraction, diffraction (8 marks / many subparts)

QP screenshot: 5054/21/M/J/25 Q5, repository PDF pages 1543–1544. MS extract: 5054_s25_ms_21, Q5.

5 **(a)** Fig. 5.1 is a diagram showing the arrangement of air particles as a longitudinal wave passes through them.

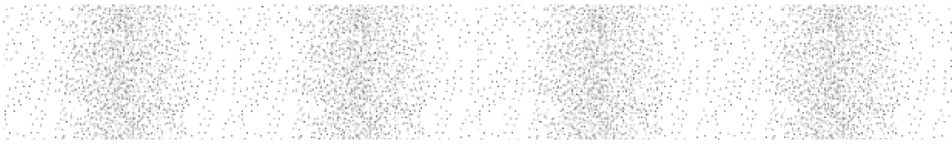

Fig. 5.1

(i) On Fig. 5.1, mark the centre of a compression with the letter C, and mark the centre of a rarefaction with the letter R. [1]

(ii) Describe the difference between a compression and a rarefaction.

....

....

.... [1]

(b) In a ripple tank, a water wave is produced by a wooden bar moving up and down on the surface of water.

(i) The wooden bar makes 45 complete oscillations in 1.0 minute.

Calculate the frequency of the wave produced.

frequency = ... Hz [1]

(ii) The frequency of the water wave is increased by moving the wooden bar up and down more quickly.

State what happens to the speed and what happens to the wavelength of the wave produced.

speed ....

wavelength ....
[2]

(iii) The crests of the water wave move into the shallow region shown in Fig. 5.2.

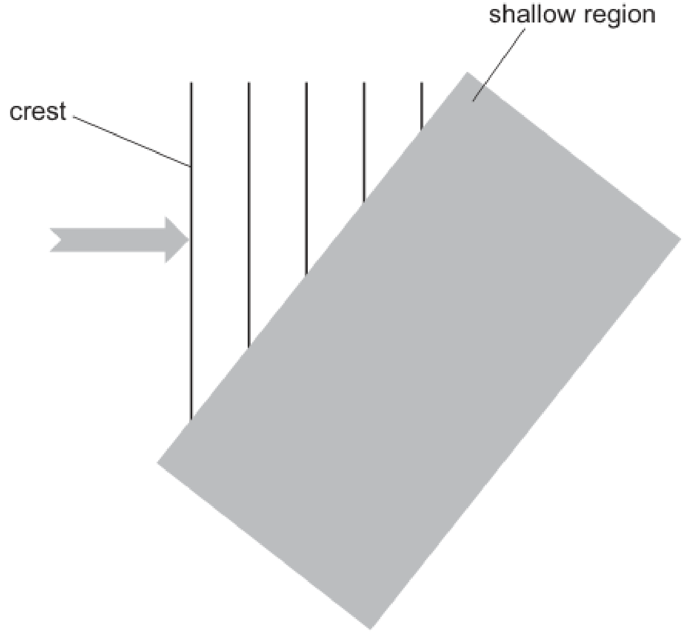

Fig. 5.2

On Fig. 5.2, draw the crests in the shallow region. [2]

(c) Describe what is meant by the diffraction of a water wave.

....

....

.... [1]

[Total: 8]

Respective mark-scheme extract

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Question 5 mark scheme extract (5054/21/M/J/25)
5(a)(i) C and R marked correctly B1
5(a)(ii) compressions are where pressure is high / particles close together / density is high B1
5(b)(i) 0.75 (Hz) B1
5(b)(ii) speed: no change B1
5(b)(ii) wavelength: decreases B1
5(b)(iii) at least 3 wavefronts in the shallow region parallel to each other and on correct side of normal M1
5(b)(iii) refracted towards the surface on correct side of surface and join wavefronts shown A1
5(c) spreading out / bending at an edge / obstacle or spreading out after passing through a (small) gap / hole B1
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2) Wave motion, frequency, wavelength, reflection in a ripple tank (8 marks)

QP screenshot: 5054/21/O/N/14 Q4, repository PDF page 880. MS extract: 5054_w14_ms_21, Q4.

4 Ripple tanks are often used to illustrate wave motion.

(a) Describe what is meant by *wave motion* in a ripple tank.

....

....

....[2]

(b) When describing wave motion, state what is meant by

(i) *frequency*,

....[1]

(ii) *wavelength*.

....

....[2]

(c) A water wave in a ripple tank strikes a barrier. Fig. 4.1 shows some wavefronts of the incident wave.

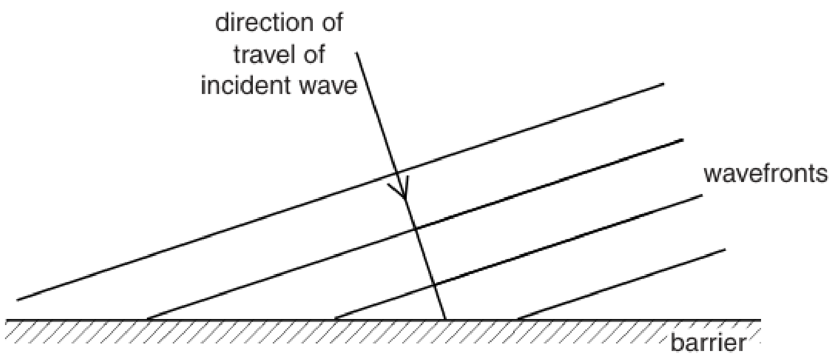

Fig. 4.1

The water wave hits the barrier and is reflected. Three of the wavefronts in Fig. 4.1 have already hit the barrier. The reflected parts of these wavefronts are not shown.

On Fig. 4.1, draw the reflected parts of these three wavefronts. [3]

Respective mark-scheme extract

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Question 4 mark scheme extract (5054/21/O/N/14)
4(a) any two from:
transmission of energy
without net movement of medium
through vibration of particles B2
4(b)(i) number of (complete) waves / cycles / oscillations per unit time / second B1
4(b)(ii) distance between (neighbouring) waves C1
distance between (neighbouring) wavefronts / points of same phase or crest to crest / trough to trough distance A1
4(c) three reflected wavefronts roughly correct direction M1
wavelengths equal to each other and incident wavelength by eye A1
reflected wavefronts joined to incident wavefronts B1
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3) Optical fibre + total internal reflection; water-wave motion and v = fλ (12 marks)

QP screenshots: 5054/22/M/J/12 Q5–Q6, repository PDF pages 2529–2530. MS extract: 5054_s12_ms_22, Q5–Q6.

5 Optical fibres are used to transmit telephone signals. Fig. 5.1 shows a ray of light th[illegible]e inside surface of an optical fibre at P.

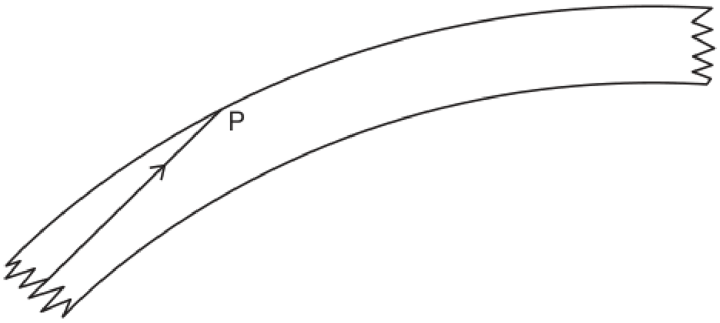

Fig. 5.1

(a) State one advantage of using optical fibres to transmit telephone signals.

....

....[1]

(b) **(i)** On Fig. 5.1, draw a normal at P and mark the angle of incidence with the letter *i*. [1]

(ii) State and explain what happens to the ray at P. Use the term *critical angle* in your answer.

....

....

....[2]

(c) The optical fibre is made of glass of refractive index 1.5.
At the start of the optical fibre, the ray enters the glass from air.
The angle of incidence in the air is 60°.

Calculate the angle of refraction in the glass.

angle = ...[2]

6 Fig. 6.1 shows a wave on the surface of water. The wave is travelling to the right.

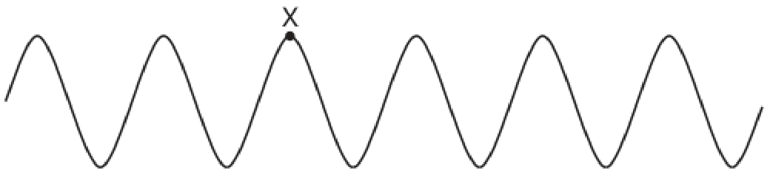

Fig. 6.1 (not to scale)

(a) Describe what is meant by *wave motion*.

....

....

....[2]

(b) On Fig. 6.1, draw an arrow to show the direction of the movement of a water molecule at X. [1]

(c) The frequency of the water wave is 2.0 Hz and the wavelength is 2.5 cm.

(i) Calculate the speed of the wave.

speed = ...[2]

(ii) On Fig. 6.1, mark a distance which shows how far a wavefront at X moves in 1.0 s. Label this distance *D*. [1]

Respective mark-scheme extract

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Questions 5–6 mark scheme extract (5054/22/M/J/12)
5(a) more telephone signals (at one time) OR greater bandwidth; more data (per sec); more signals
OR faster data/information transfer OR less attenuation; less energy/power/signal loss;
OR longer distance (before regeneration) OR more secure OR less noise/interference OR higher quality/clearer B1
5(b)(i) correct normal and angle marked B1
5(b)(ii) total internal reflection B1
angle of incidence is larger than critical angle B1
5(c) (n =) sin i / sin r in any form numerical or algebraic C1
35(.2644)° unit ° needed A1
6(a) Any 2 of:
• an oscillation/vibration/movement up and down
• carries energy
• no (net) movement of the medium/transfer of matter B2
6(b) arrow downwards or upwards or both B1
6(c)(i) (v =) fλ in any form numerical or algebraic C1
5(.0) cm/s or 0.05(0) m/s A1
6(c)(ii) line or indication labelled D of length 2 wavelengths B1
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4) Law of reflection experiment, optical fibre, plane mirror image (15 marks)

QP screenshots: 5054/22/O/N/12 Q10, repository PDF pages 1022–1023. MS extract: 5054_w12_ms_22, Q10.

10 A laser produces red light of frequency 4.7×10^{14} Hz. The speed of light in g
2.0×10^{8} m/s.

(a) Calculate the wavelength in glass of light from this laser.

wavelength = ...[2]

(b) Describe an experiment to verify *the law of reflection* for light. You may include a diagram in your answer.

....

....

....

....

....

....

....[5]

(c) Fig. 10.1 shows a ray of light travelling in an optical fibre. The ray strikes the side of the fibre at P.

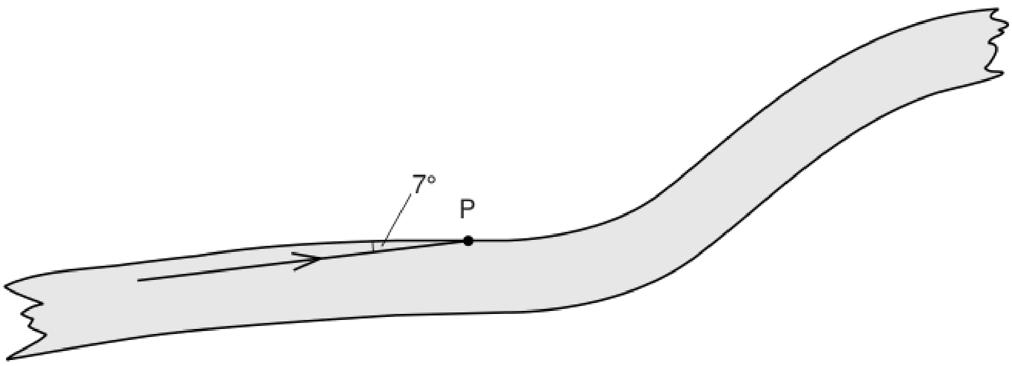

Fig. 10.1

The angle between the ray and the side of the fibre is 7°.

(i) Determine the angle of incidence of the ray at P.

angle = ...[

(ii) State and explain what happens to the ray at P.

....

....

....[2]

(d) A room is illuminated by wall lamps. Fig. 10.2 shows a mirror on the wall behind one of the lamps.

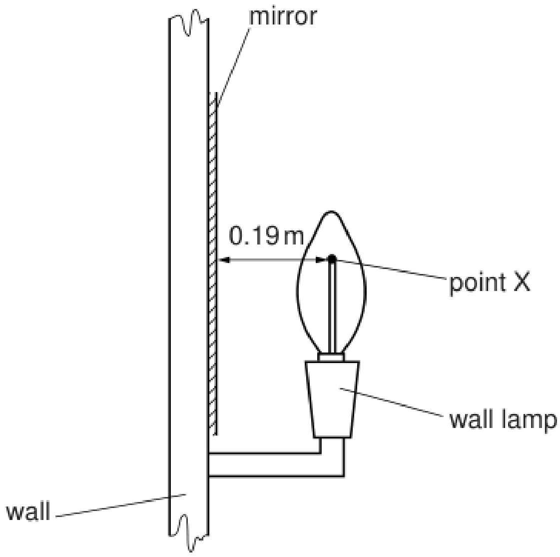

Fig. 10.2 (not to scale)

X is a point on the filament of the lamp. It is 0.19 m in front of the mirror.

(i) On Fig. 10.2, draw rays from X and locate the image of X. Label the image **I**. [3]

(ii) State the distance between **I** and the mirror.

distance = ...[1]

(iii) Suggest one advantage of placing a mirror behind the lamp in the room.

....

....[1]

Respective mark-scheme extract

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Question 10 mark scheme extract (5054/22/O/N/12)
10(a) (λ =) v/f or 2 × 10^8 / 4.7 × 10^14 C1
4.3 × 10^-7 m A1
10(b) raybox/light source/pin(s) and mirror OR laser and mirror B1
shine ray at mirror / place two pins B1
mark rays OR two more pins in line with image B1
measure i and r and equal B1
repeat B1
10(c)(i) 83° B1
10(c)(ii) total internal reflection or TIR B1
angle of incidence exceeds critical angle B1
10(d)(i) (at least) one ray from X to mirror M1
(at least) two rays from X to mirror and correct reflections A1
rays traced back to marked I or I marked in correct place (by eye) B1
10(d)(ii) 0.19 m B1
10(d)(iii) less/no light wasted or hall brighter B1
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5) Refraction by prism and semicircular glass block; Snell’s law; critical angle/TIR (15 marks)

QP screenshots: 5054/22/M/J/13 Q10, repository PDF pages 2461–2462. MS extract: 5054_s13_ms_22, Q10.

For Examiner's Use

10 A student traces the path of a ray of blue light as it enters and as it leaves a glass prism. Fig. 10.1 shows the trace obtained by the student.

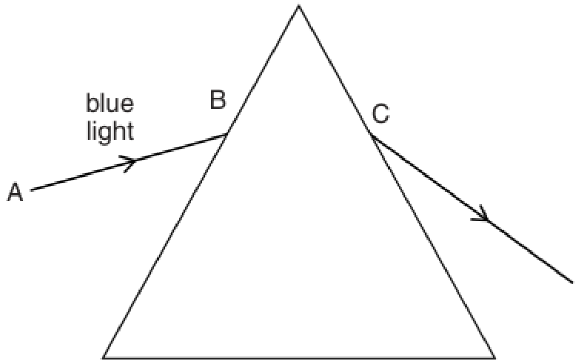

Fig. 10.1

(a) On Fig. 10.1, draw and label, at the point B, the normal, the angle of incidence *i* and the angle of refraction *r*. [3]

(b) State, in terms of the properties of light waves, why the light refracts at B.

....

.... [1]

(c) The angle of incidence for the ray of blue light at B is 45°. The refractive index of the glass is 1.5. Calculate the angle of refraction at B.

angle of refraction = ... [3]

(d) The student performs another experiment with a ray of red light along the line AB.

On Fig. 10.1, show the path taken by this ray of light as it passes through and leaves the prism. [2]

(e) The student performs another experiment with a semicircular glass block and a ray of white light. Fig. 10.2 shows the path taken by this ray of light as it enters the glass at P until it hits the straight edge at Q.

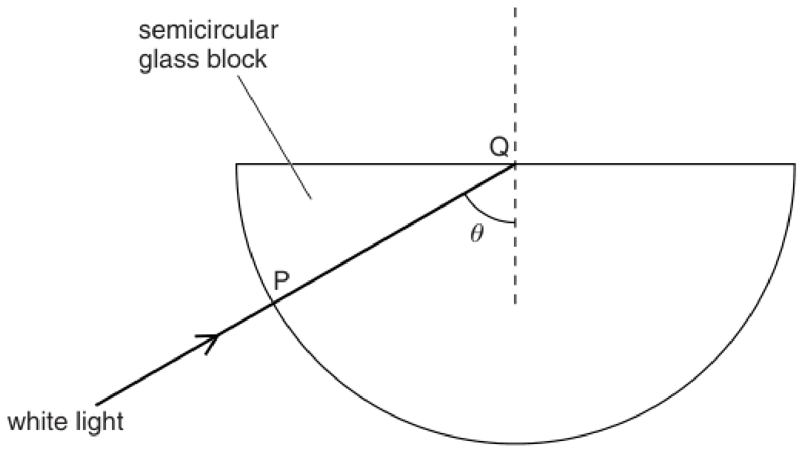

Fig. 10.2

The student finds that there is no change in direction as the ray enters the glass at P and that no light passes out of the glass at Q.

(i) Explain why the ray does not change direction at P.

....

.... [1]

(ii) Explain why no light passes out of the glass at Q.

....

....

.... [2]

(iii) On Fig. 10.2, draw the complete path followed by this ray. [1]

(iv) The student directs the ray of white light into the glass along different paths, so that the angle θ is slowly reduced.

Describe what happens to the ray at Q.

....

....

.... [2]

Respective mark-scheme extract

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Question 10 mark scheme extract (5054/22/M/J/13)
10(a) correct normal by eye B1
correct angle of incidence between candidate’s normal and incident ray B1
correct angle of refraction marked between candidate’s normal and BC B1
10(b) decrease / change in speed / wavelength B1
10(c) n = sin i / sin r seen in any form C1
(sin r =) sin 45° / 1.5 or 0.47(14) seen C1
28(.1)° C1
10(d) refracts less at first face and on correct side of normal B1
refraction at second face away from normal so that red ray and blue ray are diverging B1
10(e)(i) angle of incidence is 0 OR ray along normal/perpendicular to glass B1
10(e)(ii) angle of incidence/θ is larger than critical angle B1
total internal reflection occurs B1
10(e)(iii) reflected ray drawn correctly and emerging without refraction from block B1
10(e)(iv) (eventually) light emerges (into air at Q) OR light refracts (out at Q) OR weak refracted ray appears B1
light emerging at Q coloured in some way OR correct description of movement of reflected ray (as θ decreases) B1
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6) Diffraction through a gap; wavelength definition and v = fλ (7 marks)

QP screenshot: 5054/21/M/J/23 Q7, repository PDF page 1699. MS extract: 5054_s23_ms_21, Q7.

7 A water wave in a ripple tank diffracts as it passes through the gap in a barrier.

Fig. 7.1 shows a drawing made by a student of the crests in the pattern.

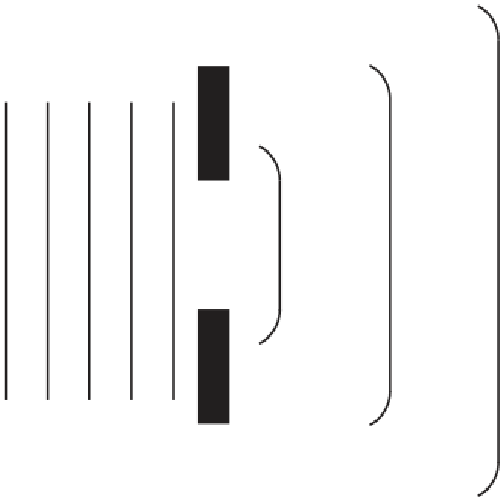

Fig. 7.1

(a) State **one** way in which the student's drawing of the crests is wrong.

....

.... [1]

(b) The gap in the barrier is now made smaller than the wavelength, as shown in Fig. 7.2.

Complete Fig. 7.2 with at least three crests to show the new diffraction pattern.

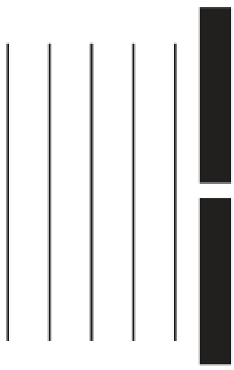

Fig. 7.2

[2]

Respective mark-scheme extract

Question 7 mark scheme extract (5054/21/M/J/23)
7(a) wavelength has altered / too large OR crests in shadow should be bent / quarter circular above or below gap
OR vertical lines too straight on two crests B1
7(b) at least 3 semicircles centered on the gap B1
wavelength constant and the same on right and left of gap and some bending B1
7(c)(i) distance between two (consecutive) identical points such as two (consecutive) crests B1
7(c)(ii) $v = f\lambda$ numerical or algebraic e.g. $f = 6/2$ (Hz) C1
3(.0) Hz A1
7(c)(iii) 1.3 cm B1

7) Transverse examples, diffraction at an edge, wavelength effect (8 marks)

QP screenshot: 5054/22/M/J/23 Q5, repository PDF page 1680. MS extract: 5054_s23_ms_22, Q5.

5 Water waves are transverse waves.

(a) Underline **two** other examples of transverse waves.

seismic P-waves **seismic S-waves** **sound** **X-rays** [1]

(b) Fig. 5.1 shows a wooden bar and a glass block in a ripple tank. The depth of water in the tank is less than the height of the glass block.

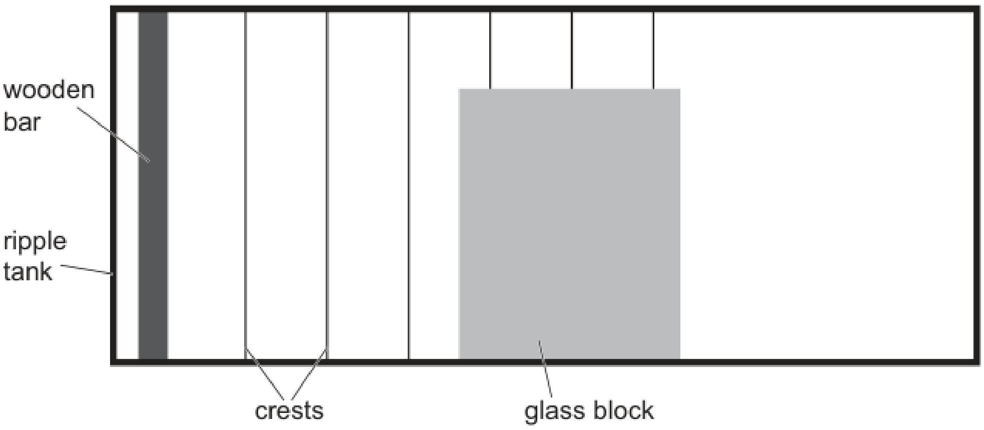

Fig. 5.1 (not to scale)

The wooden bar moves up and down once every 0.15 s to create the crests.

(i) The speed of the water wave is 27 cm/s.

Calculate the frequency and the wavelength of the wave.

frequency = ... Hz

wavelength = ... cm [3]

(ii) The wave diffracts at the right-hand edge of the glass block.

On Fig. 5.1 draw **two** crests after they pass the glass block to show the diffraction. [2]

(iii) Describe how a wave with a smaller wavelength is made with the wooden bar.

....

.... [1]

(iv) Describe how a decrease in wavelength affects the diffraction.

....

.... [1]

[Total: 8]

Respective mark-scheme extract

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Question 5 mark scheme extract (5054/22/M/J/23)
5(a) seismic S-waves and X-rays B1
5(b)(i) (wavelength =) v/f in any form OR (distance =) vt OR f = 1/T or 1/0.15 C1
frequency 6.7 Hz A1
wavelength 4.0 cm A1
5(b)(ii) straight wavefronts above block to the right with same wavelength B1
correct diffraction-wavefronts quarter circles with centre at top corner B1
5(b)(iii) increase frequency of movement of (wooden) bar B1
5(b)(iv) less diffraction or less bending B1
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